Mengdi Wang

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Education

TU Munich, *Germany, PhD* 2022.08 - present

- **Focus**: Collaborative ML, eye tracking, XR, LLM in XR
- **Thesis**: Preserving Eye Privacy in XR: Local ML Pipelines From Iris Obfuscation to Collaborative ML

ETH Zurich, *Switzerland, Master in Computer Science* 2019.09 - 2022.07

- **Thesis**: Exploration of Deep Features for Neural Style Transfer

TU Munich, *Germany, Bachelor in Informatics: Games Engineering* 2016.09 - 2019.07

- **Thesis**: Porting MLEM Algorithm for Heterogeneous Systems

Experience

TU Munich, *Chair HCTL, Research Assistant* 2022.08 - 2026.03

- Demonstrated novel federated and split learning algorithms that essentially improve model performance and convergence in heterogeneous data distribution without incurring any extra client burden or data transfer
- Developed a novel iris recognition method leveraging style features, demonstrating superior performance and robustness compared to existing approaches, and proposed the use of style transfer for iris privacy preservation
- Built real-time LLM-driven interaction system for Pepper robot (STT + TTS + joystick control), solved compatibility issues, improved user experience, and deployed in live interaction sessions with children
- Designed a plug-and-play Unity module enabling LLM-based interaction with STT and TTS for XR apps

ETH Zurich, *Computer Graphics Lab, Research Student* 2022.02 - 2022.07

- Evaluated various factors influencing the 2D-to-3D stylization process utilizing differentiable renderer
- Presented guidelines for style feature decomposition and controlling stroke granularity, and introduced a novel way for per-vertex color initialization to align reshaping and texturing outcomes

Selected Publications

- **CycleSL: Server-Client Cyclical Update Driven Scalable Split Learning**. ***M. Wang***, E. Bozkir, E. Kasneci. Proceedings of the Winter Conference on Applications of Computer Vision (WACV), 2026
- **Eye-Tracked Virtual Reality: A Comprehensive Survey on Methods and Privacy Challenges**. E. Bozkir, S. Özdel, ***M. Wang***, B. David-John, H. Gao, K. Butler,...E. Kasneci. Proceedings of the IEEE (PIEEE), 2026
- **Iris Style Transfer: Enhancing Iris Recognition with Style Features and Privacy Preservation through Neural Style Transfer**. ***M. Wang***, E. Bozkir, E. Kasneci. Proc. of the ACM on Computer Graphics and Interactive Techniques (PACMCGIT), the ACM Symposium on Eye Tracking Res. & Appl. (ETRA), 2025
- **Trade-Offs in Privacy-Preserving Eye Tracking through Iris Obfuscation: A Benchmarking Study**. ***M. Wang***, E. Bozkir, E. Kasneci. The 25th International Conference on Digital Signal Processing (DSP), 2025
- **Embedding Large Language Models into Extended Reality: Opportunities and Challenges for Inclusion, Engagement, and Privacy**. E. Bozkir, S. Özdel, K. H. C. Lau, ***M. Wang***, H. Gao, E. Kasneci. Proceedings of the 6th ACM Conference on Conversational User Interfaces (CUI), 2024
- **TurboSVM-FL: Boosting Federated Learning through SVM Aggregation for Lazy Clients**. ***M. Wang***, A. Bodonhelyi, E. Bozkir, E. Kasneci. The 38th AAAI Conference on Artificial Intelligence (AAAI), 2024

Non-Research Projects

- **CloudVis**: Interactive and dynamic visualization of large-scale atmospheric data in Unity
- **SplashyWaves**: Real-time stable fluid dynamics simulation in moving container using PBD solver
- **Kings-of-Dominia**: Multi-platform touchscreen video game replicating physics-based domino toppling

Others

- **Computer Skills**: Python (PyTorch), LM Studio, ComfyUI, Agent CLI, C# (Unity), CUDA, Linux, Git
- **Languages**: Chinese (native), English (C1), German (C1), Japanese (novice)
- **Teaching**: Human-AI Inter., Serious Games in XR, Recent Advances in Privacy, Intro to Prog. and Data Proc.